

Heart Disease Prediction (Decision Tree, KNN, and MLP)

Import Libraries:

```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPClassifier
```

Load DataSet:

```
df = pd.read_csv('/content/heart.csv')
df.head()
```

Data Preprocessing:

```
print(df.isnull().sum())
X = df.drop('target', axis=1)
y = df['target']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

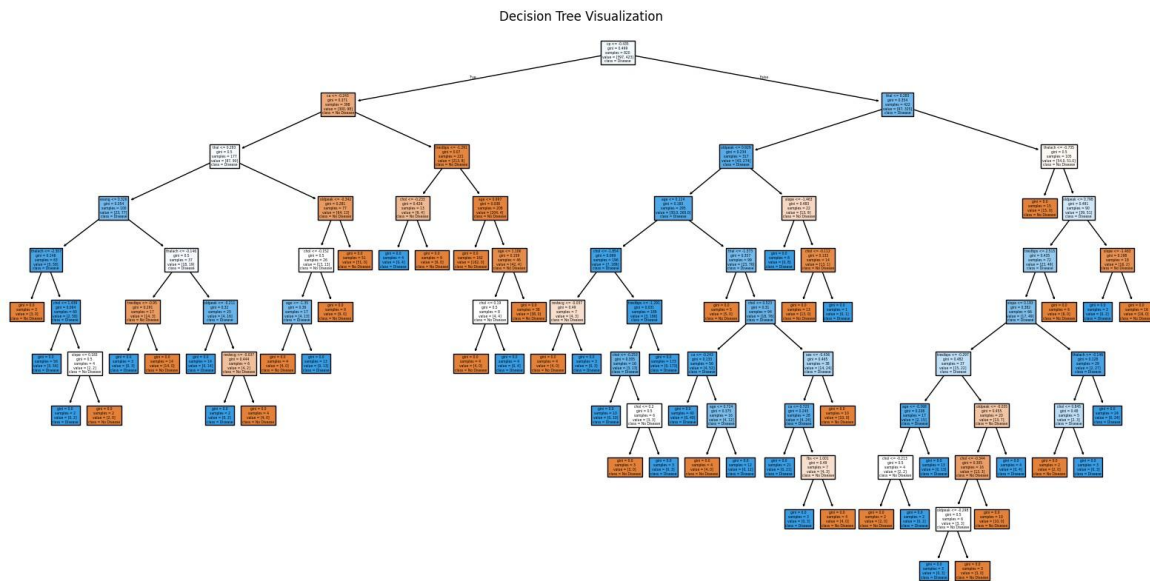
Train Models:

```
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
dt_pred = dt_model.predict(X_test)
```

```

plot_tree plt.figure(figsize=(20,10)) plot_tree(
dt_model,
    feature_names=X.columns,
    class_names=['No Disease', 'Disease'],
    filled=True
)
plt.title("Decision Tree Visualization") plt.show()

```



KNN

```
knn_model = KNeighborsClassifier(n_neighbors=5)
```

```
knn_model.fit(X_train, y_train)
```

```
knn_pred = knn_model.predict(X_test)
```

```
print("KNN Accuracy:", accuracy_score(y_test, knn_pred))
```

```
KNN Accuracy: 0.8341463414634146
```

MLP (Neural Network)

```
mlp_model = MLPClassifier(hidden_layer_sizes=(100, 50), max_iter=500,  
random_state=42) mlp_model.fit(X_train, y_train)
```

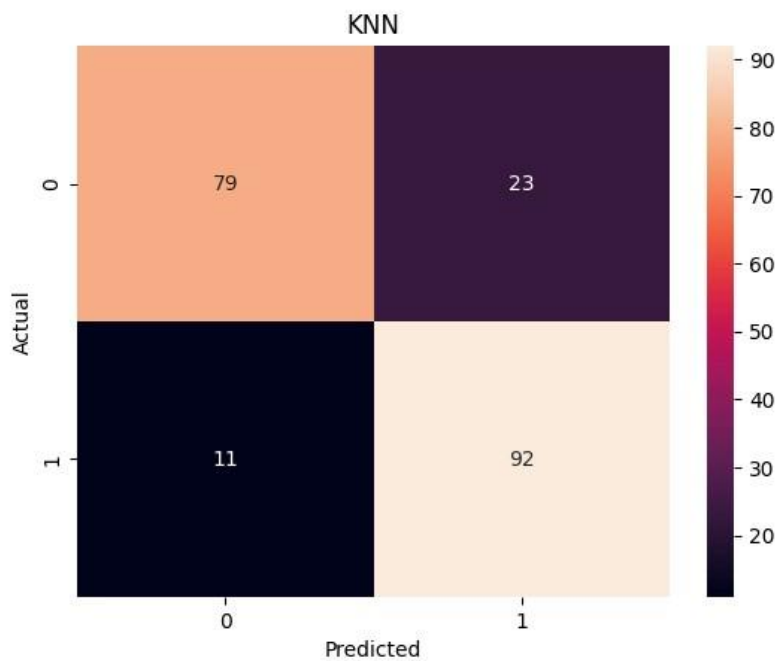
```
mlp_pred = mlp_model.predict(X_test)
```

Evaluation Function

```
def evaluate_model(name, y_test, y_pred):  
    print(f"----- {name} -----")  
    print("Accuracy:", accuracy_score(y_test, y_pred))  
    print("\nClassification Report:\n", classification_report(y_test, y_pred))  
  
    cm = confusion_matrix(y_test, y_pred)  
    sns.heatmap(cm, annot=True, fmt='d')  
    plt.title(name)    plt.xlabel('Predicted')  
    plt.ylabel('Actual')    plt.show()
```

Evaluate All Models:

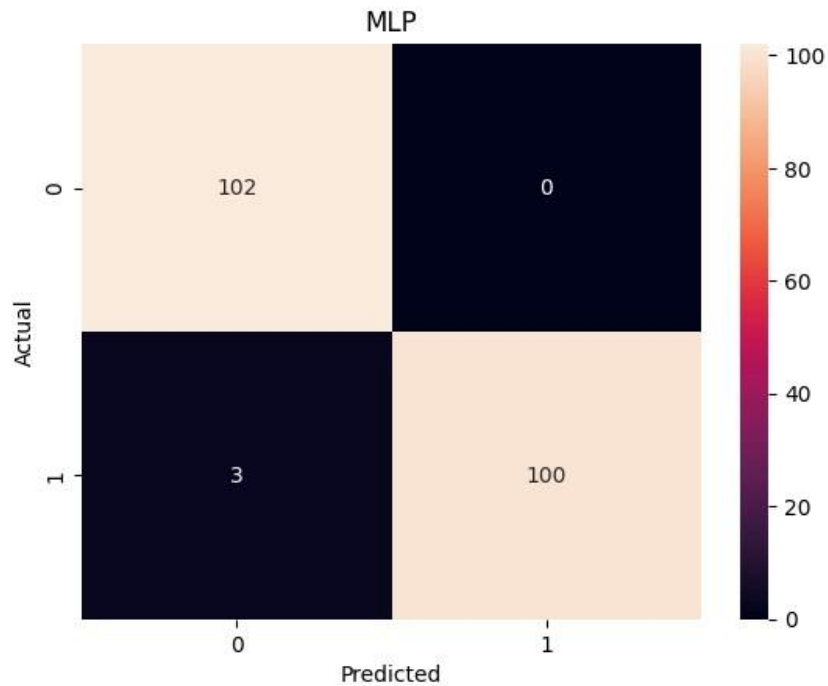
```
evaluate_model("Decision Tree", y_test, dt_pred)  
evaluate_model("KNN", y_test, knn_pred) evaluate_model("MLP",  
y_test, mlp_pred)
```



----- MLP -----

Accuracy: 0.9853658536585366

Classification Report:				precision
recall	f1-score	support		
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy				0.99
205	macro avg	0.99	0.99	
0.99	205			
weighted avg	0.99	0.99	0.99	205



Model Comparison

```
results = { "Decision Tree":
accuracy_score(y_test, dt_pred),
    "KNN": accuracy_score(y_test, knn_pred),
    "MLP": accuracy_score(y_test, mlp_pred)
}
# Convert to DataFrame results_df =
pd.DataFrame(list(results.items()), columns=['Model', 'Accuracy'])
print(results_df) # Plot comparison sns.barplot(x='Model',
y='Accuracy', data=results_df) plt.title("Model Comparison")
plt.show()
```

	Model	Accuracy
0	Decision Tree	0.985366
1	KNN	0.834146
2	MLP	0.985366

